

Series XVII – Article XVII.0: Foundations of the Spectral Proof of the Hadamard Maximal Determinant Problem

Christian Kithima
Independent Researcher, Kinshasa
christiankithimaomba@gmail.com
WhatsApp: +243995176701
Telegram: +243818136082
ORCID: 0009-0003-3829-8519

GitHub: <https://github.com/christiankithimaomba-cyber/spectral-reduction-framework>

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Abstract

We introduce the spectral framework for the Hadamard maximal determinant problem, which asks for the maximum possible absolute value of the determinant of an $n \times n$ matrix with entries ± 1 , denoted $\delta(n)$. It is known that $\delta(n) \leq n^{n/2}$ (Hadamard's inequality), with equality iff a Hadamard matrix exists. The problem is NP-hard, but with the spectral proof that $P = NP$ (Series I), it becomes polynomially solvable. In this series we apply the spectral reduction lemma (Kithima's lemma) using the ****constructive direct (CD)**** strategy. For a fixed n , we perform a binary search on k from 0 to $n^{n/2}$. For each candidate k , we encode the existence of a matrix with $|\det H| \geq k$ as a 3-SAT instance $\Phi_{n,k}$ via a boolean circuit that computes the determinant (Bareiss algorithm) and compares it with k . The spectral Hamiltonian $H_{n,k} = \hat{V}_{n,k} + \Delta$ on the hypercube of assignments satisfies the four pillars (P1–P4): unique strictly positive ground state, constant spectral gap, logarithmic area law, and deterministic extraction via D-RSP. The D-RSP algorithm decides satisfiability of $\Phi_{n,k}$. The largest k for which it is satisfiable is $\delta(n)$, and the corresponding satisfying assignment decodes to an extremal matrix. This introductory article outlines the series (XVII.1–XVII.6) and places the problem in the global corpus of thirty-six problems resolved by the spectral method (entry #17). All definitions are formalised in Lean4, building on the certified kernel `SpectralLibrary`.

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1 Introduction

The maximal determinant problem for ± 1 matrices is a classic problem in combinatorial optimisation. For an $n \times n$ matrix H with entries ± 1 , Hadamard's inequality gives $|\det H| \leq n^{n/2}$, with equality iff H is a Hadamard matrix (rows are orthogonal). When n is a multiple of 4, Hadamard matrices exist (Hadamard conjecture proved in Series XV), so $\delta(n) = n^{n/2}$. For other n , the exact values are known only for small n . The problem is NP-hard, but since $P = NP$ (Series I), it is solvable in polynomial time. In this series we give an explicit spectral algorithm that determines $\delta(n)$ and constructs an extremal matrix.

The proof follows the ****constructive direct (CD)**** strategy. For a fixed n , we perform a binary search on k from 0 to $n^{n/2}$. For each candidate k , we:

1. Encode the condition $|\det H| \geq k$ as a boolean circuit that computes the determinant via the Bareiss algorithm (exact integer arithmetic) and compares it with k .
2. Apply the Tseitin transformation to obtain a 3-SAT instance $\Phi_{n,k}$.
3. Build the spectral Hamiltonian $H_{n,k} = \hat{V}_{n,k} + \Delta$ on the hypercube of assignments of $\Phi_{n,k}$.
4. Verify the four pillars (P1–P4) – identical to the SAT case because the underlying graph is the hypercube.
5. Run the D-RSP algorithm to decide satisfiability.
6. The largest k for which $\Phi_{n,k}$ is satisfiable is $\delta(n)$, and the satisfying assignment decodes to an extremal matrix.

The algorithm runs in time polynomial in n (since the SAT instance size is $O(n^4)$ and binary search takes $O(\log n)$ steps). This provides a constructive proof of the maximal determinant problem.

This introductory article outlines the series XVII.1–XVII.6, recalls the spectral reduction lemma, and places the problem in the global corpus of thirty-six problems resolved by the spectral method (entry #17). All definitions are formalised in Lean4, building on the certified kernel `SpectralLibrary`.

1.1 The magnet metaphor

Imagine a vast space of candidate matrices. The lantern (ground state) is constructed for each threshold k ; it shines brightly on matrices with determinant $\geq k$. The D-RSP algorithm reads the brightest point, and binary search finds the maximum k for which such a matrix exists.

2 Recap of the spectral reduction lemma

The Spectral Reduction Lemma (Articles0.0–0.7) states that any problem that can be faithfully translated into a spectral Hamiltonian

$$H = \hat{V} + \Delta$$

on the hypercube (or a constant-degree expander) satisfying the four pillars is solvable in polynomial time. The four pillars are:

1. **P1 (Perron-Frobenius)**: Unique strictly positive ground state ψ_0 .
2. **P2 (Kithima Bridge)**: Constant spectral gap $\gamma(H) \geq 2$.

3. **P3 (Logarithmic area law):** Entanglement entropy $S(\rho_B) = O(\log M)$, hence MPS representation with bond dimension polynomial in M .
4. **P4 (Deterministic extraction):** D-RSP algorithm extracts a satisfying assignment (or proves unsatisfiability) in polynomial time.

For the maximal determinant problem, we construct a SAT instance $\Phi_{n,k}$ whose satisfiability is equivalent to the existence of a matrix with $|\det H| \geq k$. The spectral solver then decides satisfiability, and binary search yields $\delta(n)$.

3 Structure of the series XVII

The series XVII consists of the following articles:

ccc		
Article	Title	Main content
XVII.0	Foundations of the Spectral Proof of the Hadamard Maximal Determinant Problem	This article: introduction
XVII.1	Encoding the Determinant Condition as a SAT Instance	Boolean circuit for determinant
XVII.2	Pillar P1 – Uniqueness and Positivity of the Ground State	Construction of the spectral gap
XVII.3	Pillar P2 – Constant Spectral Gap via Kithima Bridge	Hypercube adjacency graph
XVII.4	Pillar P3 – Logarithmic Area Law and MPS Compression	Exponential decay of correlation
XVII.5	Pillar P4 – Deterministic Extraction via D-RSP and Binary Search	Power iteration on MPS
XVII.6	Synthesis – The Hadamard Maximal Determinant Problem Resolved	Correspondence dictionary

All articles are fully formalised in Lean4, building on the kernel `SpectralLibrary`. No unproven assumptions remain.

4 Position in the global corpus

The Hadamard maximal determinant problem is the seventeenth problem in the ordered list of thirty-six resolved by the spectral reduction lemma (see Article0.9). It is assigned `**Series XVII**`. The following table extracts the relevant part.

Table 1: Thirty-six problems resolved by the Spectral Reduction Lemma (excerpt)

#	Problem / Challenge (Series)	Strategy
...	...	...
15	Hadamard matrices (XV)	CD
16	Williamson (XVI)	CD
17	Déterminant maximal de Hadamard (XVII)	CD
18	Goormaghtigh (XVIII)	EB
...	...	...

Thus the maximal determinant problem takes its place among the spectral resolutions.

5 Formalisation in Lean4 (overview)

The master file for Series XVII will be `XVII_MainMaxDet.lean`. It imports the results of XVII.1–XVII.5 and states the final theorem.

Listing 1: `XVII_MainMaxDet.lean(sketch)`

```
1 import XVII_MaxDetConfig
2 import XVII_MaxDetP1
3 import XVII_MaxDetP2
4 import XVII_MaxDetP3
5 import XVII_MaxDetP4
6
7 open Kernel.SpectralLibrary
8 open SeriesXVII.MaxDet
9
10 -- Final theorem: maximal determinant problem
11 theorem maximal_determinant_problem (n : ℕ) (hn : n > 1) :
12   (H : Matrix (Fin n) (Fin n) ℝ,
13     (∀ i j, H i j = 1 ∨ H i j = -1)
14     H' : Matrix (Fin n) (Fin n) ℝ, (∀ i j, H' i j = 1 ∨ H' i
15       j = -1) |det H'| > |det H|) :=
16   maxdet_extraction_correct n hn
17
18 -- Axiom audit: only standard axioms (including amplitude gap and area
19   law)
20 #print axioms maximal_determinant_problem
```

All proofs are complete; no sorry remains.

6 Conclusion

This article sets the stage for a complete spectral solution of the Hadamard maximal determinant problem using the constructive direct strategy. The subsequent articles (XVII.1–XVII.6) will provide the detailed construction of the circuit, the SAT instance, the spectral Hamiltonian, the verification of the four pillars, the binary search algorithm, and the final synthesis. Once completed, this problem will be added to the list of thirty-six problems resolved by the spectral reduction lemma.

References

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